Projects
Inbox 7
Model
Rules
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Revisions

Projects 2

New project

Demo ship IFC4

Rev A → Rev B

Passenger ship · > 36 passengers | Fire & life safety · Accessibility · Class (To verify)

LAST RUN · DECK 5 · REV B · 14:36

3 Fail | 1 Warning | 1 Cannot verify | 87 Pass | 4 Dormant

96 checks on the architecture and outfitting model. HVAC model: 1 Fail (H-01).

2 decisions waiting in the Fire & life safety inbox · BCF sonariq_rev-b.bcfzip

Open inbox | Open model | Rev A vs Rev B

RCRV Navisworks → IFC

Rev 1

Research vessel · 28 on board | 16 scientists · 12 crew | Units: in

LAST RUN · REV 1 · ACCESSIBILITY SAMPLE

D-02-114 · 30 in | D-02-118 · 34 in | SOLAS > 36 passengers

Applicability set from the vessel profile: rules written for ships carrying more than 36 passengers are listed as Not applicable, not hidden.

Import mapping has **3 rows to review** and **1 unmapped**. Checks that depend on them show Cannot verify.

Open model | Review mapping

Drop files to add or update a model

IFC4 | OCX · .3docx | BCF · .bcfzip

IFC4 for every discipline. OCX for hull structure, optional. BCF brings issues in from other design tools. Files stay on your servers.

Choose files

RECENT FILES — Demo ship · RCRV

sonariq_rev-b.bcfzip	written · 6 topics
Demo_ship_RevB.ifc	Checked
RCRV_converted.ifc	Mapping to review
Demo_ship_HVAC_RevB.ifc	validating · 64%
hull_block_12.3docx	Not read

hull_block_12.3docx: OCX schema version not recognised. Export OCX 3.0 or skip; IFC structure still checks.

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Settings

Data stays on-premises
v0.1 · prototype

Project Demo ship ▾ Revision Rev B ▾ OR-117 · 14:36

Find F-07, FD-5-165P, II-2/13…

Units mm in

MS Mikko Saari
Fire & life safety lead

Inbox

Demo ship · Rev B · sent from FreeCAD 14:43 · sonariq_rev-b.bcfzip

Fire & life safety 5 | Structure 0 | Architecture 0 | HVAC 1 | Systems 0 | Accessibility 1

All open | Fail | Warning | Cannot verify | Accepted | Resolved

Waiting for me

Sort: Decision needed first ▾

1 selected — Assign… Export BCF Clear

Approve and decline are one at a time, so each one is signed.

STATUS	FINDING	LOCATION	RULE	OWNER	AGE	DECISION
Pass	F-07 Stair 3B · fix 2 applied in FreeCAD	Stair 3B · Decks 4–7	FSS Code Ch.13 §2.1.2	Aino Laine	33 min	Approve fix
Fail	F-08 Dead-end corridor · accepted with reason	Deck 5 · Fr 146–150 · P	SOLAS II-2/13.3.1.2	Aino Laine	24 min	Awaiting your approval
Fail	F-01 Fire door FD-5-165P below bulkhead class	Deck 5 · Fr 165 · P	SOLAS II-2/9.2.2.1.1	Aino Laine	45 min	Open
Fail	F-09 Corridor pinch · 850 mm, required 900 mm	Deck 5 · Fr 152 · S	FSS Code Ch.13	Aino Laine	33 min	Open
Cannot verify	U-01 Door D-5-171-S has no fire class	Deck 5 · Fr 171 · S	SOLAS II-2/9 · door fire integrity	Aino Laine	45 min	Needs data

Click a finding to open its thread. Times are local (EET). Now 15:05.

Pass Stair 3B · fix 2 applied

Fix applied · needs your approval

Fail F-07 Rule engine

Stair 3B is narrower than the people it serves need

Stair 3B · Decks 4–7 · Deck 5

FS Fire & life safety

FSS Code Ch.13 §2.1.2 sample rule text

> Stairway clear width shall not be less than W, worked out from the persons served on each deck the stair connects.

REQUIRED	MEASURED	MARGIN
1,380 mm	1,350 mm	−30 mm

FORMULA · INPUTS FROM THE MODEL

$$W = (N_1 + N_2 + 0.5\,N_3 + 0.25\,N_4) \times 10\ \text{mm}$$
$$= (\ 68_{N_1\ D5} + 44_{N_2\ D6} + 0.5 \cdot 40_{N_3\ D4} + 0.25 \cdot 24_{N_4\ D7}\ ) \times 10$$
$$= (68 + 44 + 20 + 6) \times 10 = 138 \times 10 = \mathbf{1{,}380\ mm}$$

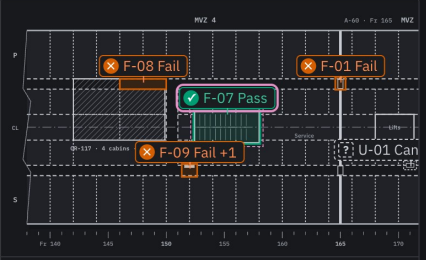
Highlighting in 3D: Deck 5 · 68 persons: 60 before OR-117, + 8 berths in the four new cabins.

Rev A needed 1,300 mm (N_1 was 60). The stair spans four decks nobody touched.

HOW IT WAS MEASURED

Clear width between handrails, from IFC geometry. Persons N from berths of the IfcSpaces each deck sends to Stair 3B.

IfcStair Stair-3B · checked Rev B · 14:32



3D snapshot · Deck 5 · Rev B · 14:36 Open in model viewer

THREAD system events, decisions and comments, newest last

SQ **Rule engine** found F-07 after OR-117 (cabin block × 4, 8 berths) 14:32 · Rev B
Fail required 1,380 mm · measured 1,350 mm · margin −30 mm Rule engine

AL **Aino Laine** Outfitting designer applied fix 2 from FreeCAD 14:36 · Rev B
> Keep four cabins, one berth each. N_1 68 → 64 on Deck 5. Change order OR-117.

SQ **Rule engine** re-checked 14:36 · Rev B
Pass 1,340 ≤ 1,350 mm · margin 10 mm Rule engine

AL **Aino Laine** Outfitting designer sent to review 14:43 · Rev B
> Margin is 10 mm. Any new berth on Decks 4–7 will reopen this.

Comment

Add a comment. It is signed with your name, role, time and revision.

Approve | Request change | Decline

Signing as Mikko Saari · Fire & life safety lead · Rev B

SIGNED HISTORY

- **Rule engine** · deterministic Fail · 1,380 / 1,350 mm 14:32 Rev B
- **Aino Laine** · Outfitting designer applied fix 2 14:36 Rev B
- **Rule engine** · deterministic Pass · 1,340 / 1,350 mm 14:36 Rev B

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Project Demo ship · Revision Rev B · OR-117 · 14:36

Find F-07, FD-5-165P, II-2/13…

Units mm in

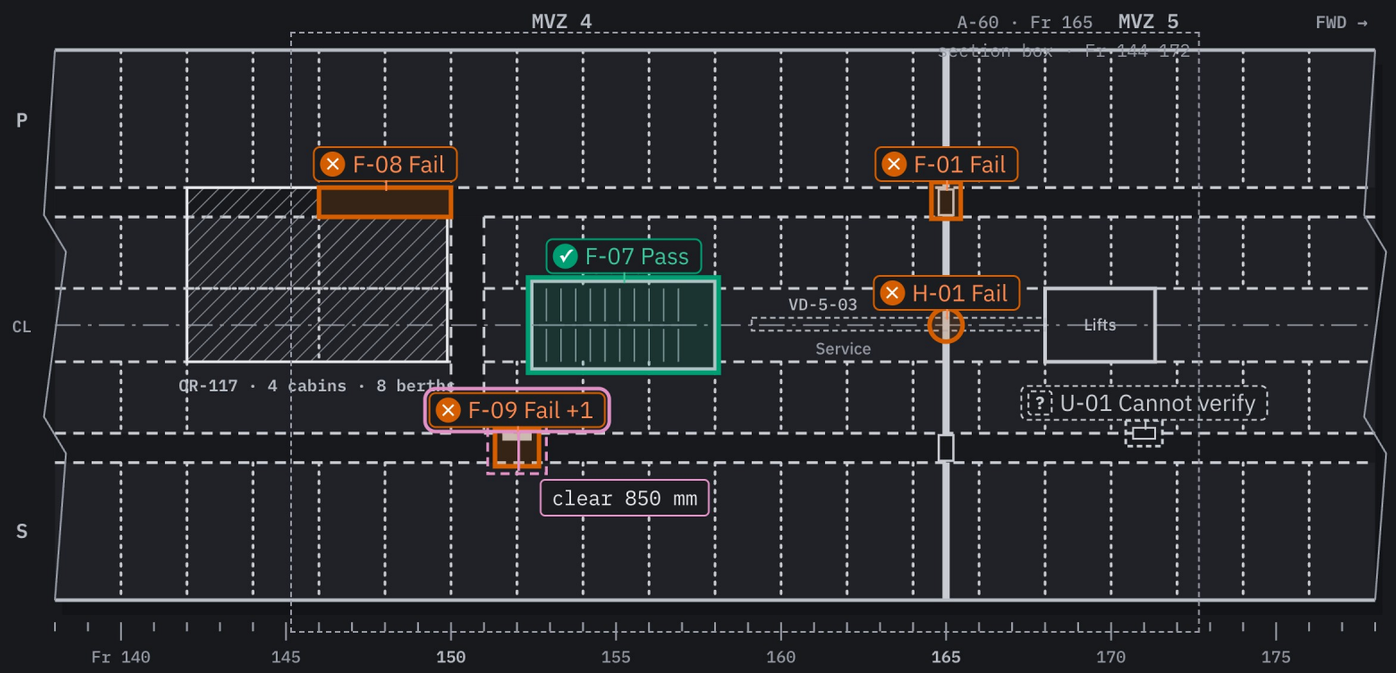
MS Mikko Saari
Fire & life safety lead

Deck Deck 4 | Deck 5 | Deck 6 | Deck 7

Layers ☑ Architecture ☐ Structure ☑ HVAC ☐ Systems ☑ Findings

Orbit | Measure | Section box | Fit | Top

Demo ship · Deck 5 · top view · Rev B · measure: Fr 152 · S



A-60
A-0
B-15
B-0

DECK STACK

Deck 7
Deck 6
Deck 5
Deck 4

Stair 3B · Decks 4–7 · Pass

FINDINGS ON THIS DECK 7

Fail — F-01 FD-5-165P below bulkhead class
A-0 fitted · A-60 required

Fail — F-08 Dead-end corridor
4.2 m deep · 1.2 m wide · awaiting approval

Fail — F-09 Corridor pinch
850 mm · required 900 mm

Fail — H-01 VD-5-03 without fire damper
crosses A-60 MVZ bulkhead · Fr 165

Warning — W-01 Corridor below guideline
850 mm · guideline 915 mm (36 in)

? Cannot verify — U-01 D-5-171-S has no fire class
add the property or confirm

Pass — F-07 Stair 3B
1,340 ≤ 1,350 mm · fix 2

Click a pin or a row to select. Measure shows clear width between finished surfaces.

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v0.1 · prototype

Project RCRV · Revision Rev 1 · Navisworks → IFC

Find F-07, FD-5-165P, II-2/13…

Units mm | in

AL Aino Laine
Outfitting designer

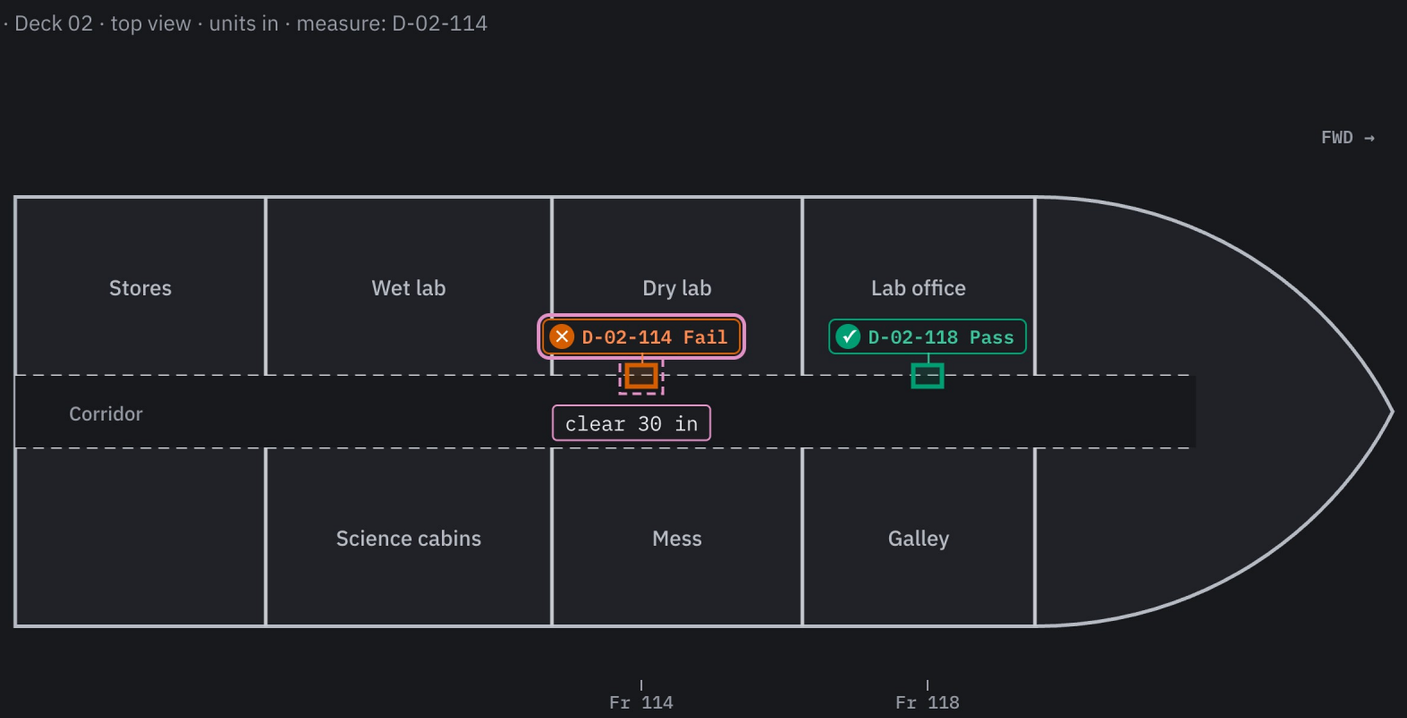
Applicability · research vessel · 28 people on board 16 scientists, 12 crew · SOLAS rules for ships carrying more than 36 passengers: Not applicable — Change vessel profile

Deck: Deck 01 | **Deck 02** | Deck 03

Layers: ☑ Architecture ☐ Structure ☐ HVAC ☐ Systems ☑ Findings

Orbit | Measure | Section box | Fit | Top

RCRV · Deck 02 · top view · units in · measure: D-02-114



A-60
A-0
B-15
B-0

Geometry is a converted Navisworks mesh. Door widths are measured from the mesh; 3 property mappings still need review.

FINDINGS ON THIS DECK 2

Fail — D-02-114 **Door clear width**
30 in · required 32 in (762 / 813 mm) · ADA 2010 §404.2.3

Pass — D-02-118 **Door clear width · control**
34 in ≥ 32 in (864 / 813 mm)

NOT APPLICABLE TO THIS VESSEL

Not applicable
SOLAS II-2/9.2.2.1.1
MVZ bulkhead and door integrity · passenger ships > 36 passengers

Not applicable
SOLAS II-2/13.3.1.2
Dead-end corridors · passenger ships > 36 passengers

Not applicable
FSS Code Ch.13 §2.1.2
Stair width from persons served · passenger ships > 36 passengers

Rules are listed as Not applicable with the reason, never hidden. Change the vessel profile to re-run.

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Import mapping review

RCRV · RCRV_converted.ifc · source: Navisworks layers and item names

All 12 | Matched 8 | Needs review 3 | Unmapped 1

J K move · A accept · E edit · U leave unmapped

Accept all matched

STATE	SOURCE · LAYER / ITEM	IFC CLASS	DECK	FRAME	SIDE	MVZ	SOLAS CATEGORY	BERTHS
Matched	A-DOOR / Door_Single_30in	IfcDoor	02	114	P	—	—	—
Matched	A-DOOR / Door_Single_34in	IfcDoor	02	118	P	—	—	—
Matched	A-WALL-FIRE / Bhd_A60	IfcWall · A-60	02	110	CL	—	—	—
Matched	A-WALL / Partition_B15	IfcWall · B-15	02	—	—	—	—	—
Needs review	A-SPACE / LAB_WET	IfcSpace	02	106–112	P	—	?	—
Needs review	A-SPACE / CABIN_SCI_01	IfcSpace	02	108–110	S	—	Accommodation	?
Matched	A-STAIR / Stair_Main	IfcStair	01–03	120	CL	—	—	—
Matched	M-DUCT / Duct_Supply	IfcDuctSegment	02	—	—	—	—	—
Unmapped	0 / Solid 4471	—	?	?	?	—	—	—
Needs review	A-DOOR / property FR = “60”	IfcDoor.FireRating	02	—	—	—	—	—
Matched	S-DECK / Deck_02_Plate	IfcSlab	02	—	—	—	—	—
Matched	A-SPACE / CORR_02_MAIN	IfcSpace	02	104–124	CL	—	Corridor	—

Needs review — A-DOOR / property FR = “60”

The source stores a number, not a fire class. “60” could mean A-60 or a 60-minute rating from another standard.

Seen on: 7 doors on Deck 02
Also seen: “0”, “15”, empty
Suggestion: FR 60 → A-60, FR 0 → A-0 (layer A-WALL-FIRE)

MAP FR “60” TO

- (●) FireRating = A-60
- ( ) FireRating = B-15
- ( ) Leave unmapped (checks show Cannot verify)

Apply to 7 doors | Next ↓

CANNOT VERIFY UNTIL MAPPED

- Door fire integrity on Deck 02 · needs FR mapping
- Stair width from persons served · needs berths on CABIN_SCI_*
- Fire risk by space category · needs a category for LAB_WET
- Anything touching Solid 4471 · unmapped geometry

Mapping rules are saved per project and reused on the next import. Matched rows are not re-asked.

Rulebook

Every verdict comes from these rules. Rule text shown here is sample text.

Fire & life safety Verified
SOLAS II-2 · FSS Code Ch.13
5 rules in this sample

Accessibility Verified
PVAG 2013 draft · ADA 2010
2 rules in this sample

Class · DNV To verify
Sample references
2 rules in this sample

ID	CLAUSE	WHAT IT CHECKS	APPLIES TO	PARAMETERS	STATUS	TES
R-FLS-01	SOLAS II-2/9.2.2.1.1	A door in an MVZ bulkhead has at least the bulkhead's fire class.	Passenger ship · > 36 passengers	bulkhead class · door FireRating	Verified	
R-FLS-02	SOLAS II-2/13.3.1.2	A dead-end recess is no deeper than it is wide.	Passenger ship · > 36 passengers	depth m · width m	Verified	
R-FLS-03	FSS Code Ch.13 §2.1.2	Stair clear width covers the persons served on each deck.	Passenger ship · > 36 passengers	N_1…N_4 · width mm	Verified	
R-FLS-04	FSS Code Ch.13	Escape corridor clear width at least 900 mm.	Passenger ship · > 36 passengers	clear width ≥ 900 mm	Verified	
R-FLS-05	SOLAS II-2/9.7	A duct through an A-class division carries a fire damper.	All ship types in profile	duct · division class · damper	Verified	
R-ACC-01	PVAG 2013 draft	Corridor clear width guideline 915 mm (36 in). Draft source, so a warning, never a fail.	Passenger ship	clear width ≥ 915 mm	Verified	
R-ACC-02	ADA 2010 §404.2.3	Door clear width at least 32 in with the door open 90°.	Any vessel with accessibility on	clear width ≥ 32 in	Verified	
R-DNV-01	DNV [clause to confirm]	Class rule placeholder. Runs, but every verdict carries To verify.	Per class notation	[to confirm]	To verify	
R-DNV-02	DNV [clause to confirm]	Class rule placeholder. Runs, but every verdict carries To verify.	Per class notation	[to confirm]	To verify	

R-FLS-03 FSS Code Ch.13 §2.1.2

Verified

Stair clear width must cover the people the stair serves, weighted by deck.

$$W = (N_1 + N_2 + 0.5\,N_3 + 0.25\,N_4) \times 10\ \text{mm}$$

pass if clear width ≥ W

Applies to	Passenger ship · > 36 passengers
Inputs	N_1…N_4 persons per deck · stair clear width between handrails
Units	mm, as written in the rule
Cannot verify if	berths missing on a served deck, or handrails not modelled
Tests	[n] cases incl. Demo ship Rev A (1,300 mm) and Rev B (1,380 mm)
Reviewed	[name · date]

Open tests | History

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Benchmark

Planted violations and clean controls. Results are written by the run, never by hand.

Run benchmark

PLANTED CASES	CAUGHT	MISSED	FALSE ALARMS	RUN DATE
8	[after run]	[after run]	[after run]	[after run]
+ 2 clean controls	of 8 planted	planted, not found	findings with no planted case	engine version [after run]

#	RULE	LOCATION	WHAT WAS PLANTED	EXPECTED	FOUND
1	SOLAS II-2/9.2.2.1.1	Deck 5 · Fr 165 · P	FD-5-165P fitted A-0 in the A-60 MVZ 4/5 bulkhead	Fail	[after run]
2	FSS Code Ch.13 §2.1.2	Stair 3B · Decks 4–7	Rev B: 1,350 mm against 1,380 mm required	Fail	[after run]
3	SOLAS II-2/13.3.1.2	Deck 5 · Fr 146–150 · P	Dead-end 4.2 m deep, 1.2 m wide	Fail	[after run]
4	FSS Code Ch.13	Deck 5 · Fr 152 · S	Corridor 850 mm against 900 mm	Fail	[after run]
5	PVAG 2013 draft	Deck 5 · Fr 152 · S	Corridor 900 mm against 915 mm guideline	Warning	[after run]
6	SOLAS II-2/9 · door fire integrity	Deck 5 · Fr 171 · S	D-5-171-S with no fire class property	Cannot verify	[after run]
7	SOLAS II-2/9.7	Deck 5 · Fr 165	VD-5-03 through the A-60 bulkhead, no damper	Fail	[after run]
8	ADA 2010 §404.2.3	RCRV · Deck 02 · Fr 114	D-02-114 clear width 30 in against 32 in	Fail	[after run]
C1	ADA 2010 §404.2.3	RCRV · Deck 02 · Fr 118	Control: D-02-118 at 34 in	Pass	[after run]
C2	SOLAS > 36 passengers rules	RCRV · whole model	Control: research vessel, 28 on board	Not applicable	[after run]

A case counts as caught only if the engine returns the expected status for the expected element and rule. A finding on any other element is a false alarm. Controls catch rules that fire when they should not.

Rev A → Rev B

change order OR-117 · cabin block × 4 · Deck 5 · aft of Fr 150 · confirmed by Aino Laine 14:32

Rev A | Rev B

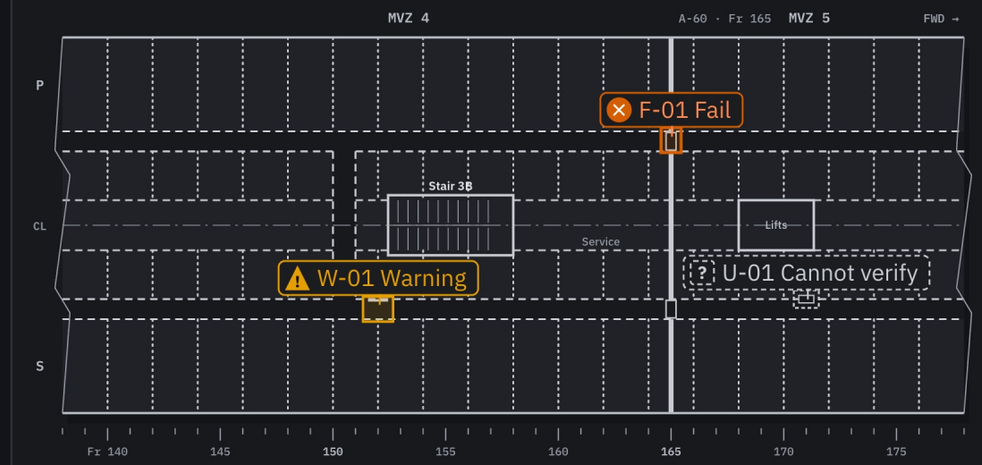
CHANGED ELEMENTS
4 cabins added
+ 2 corridors reshaped

CHECKS RE-RUN
14
of 96 · only what the change touches

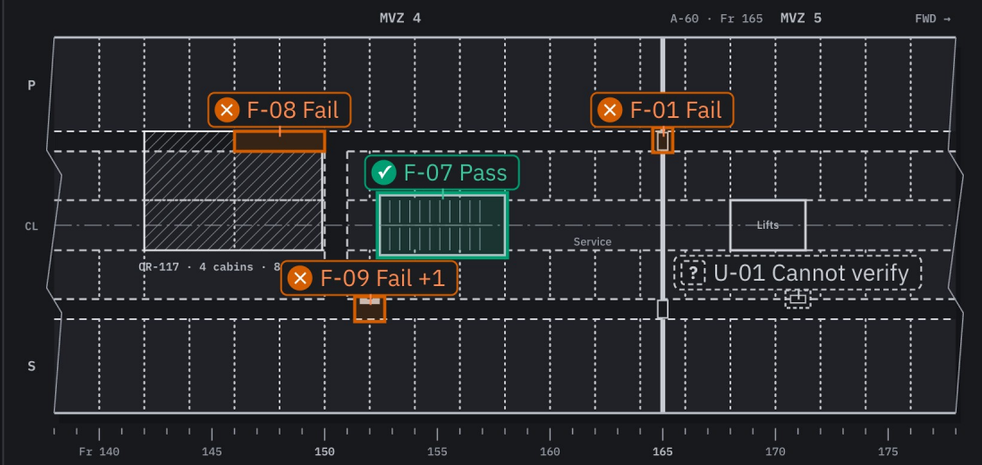
VERDICTS CHANGED
3
at 14:32 · F-07 back to Pass at 14:36

BERTHS ON DECK 5
60 → 64
68 at 14:32, 64 after fix 2

Rev A · 14:20 — 1 fail · 1 warning · 1 cannot verify



Rev B · 14:36 · hatched = new — 3 fail · 1 warning · 1 cannot verify



VERDICTS THAT CHANGED

F-07	Stair 3B · Decks 4–7	Pass → Pass
	1,300 → 1,380 mm Fail at 14:32 → 1,340 mm Pass at 14:36 (fix 2)	
F-08	Dead-end · Fr 146–150 · P	Pass → Fail
	depth 4.2 m > width 1.2 m · accepted with reason, awaiting approval	
F-09	Corridor pinch · Fr 152 · S	Pass → Fail
	clear width 900 → 850 mm · required 900 mm	
W-01	Corridor guideline · Fr 152 · S	Warning → Warning
	900 → 850 mm · guideline 915 mm · status unchanged	

CHANGED ELEMENTS

Cabin block × 4 — added
Deck 5 · aft of Fr 150 · 8 berths at 14:32 · 4 berths after fix 2 · OR-117

COR-5-146-P — reshaped
closed at Fr 146 · open end at Fr 150 · now a dead-end

COR-5-152-S — reshaped
clear width 900 → 850 mm at Fr 152

Stair 3B — not edited
geometry unchanged · N_1 60 → 68 → 64 · re-checked because Deck 5 berths changed

SonarIQ

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Project Demo ship ▾ Revision Rev B ▾ OR-117 · 14:36

Find F-07, FD-5-165P, II-2/13…

Units mm in

MS Mikko Saari
Fire & life safety lead